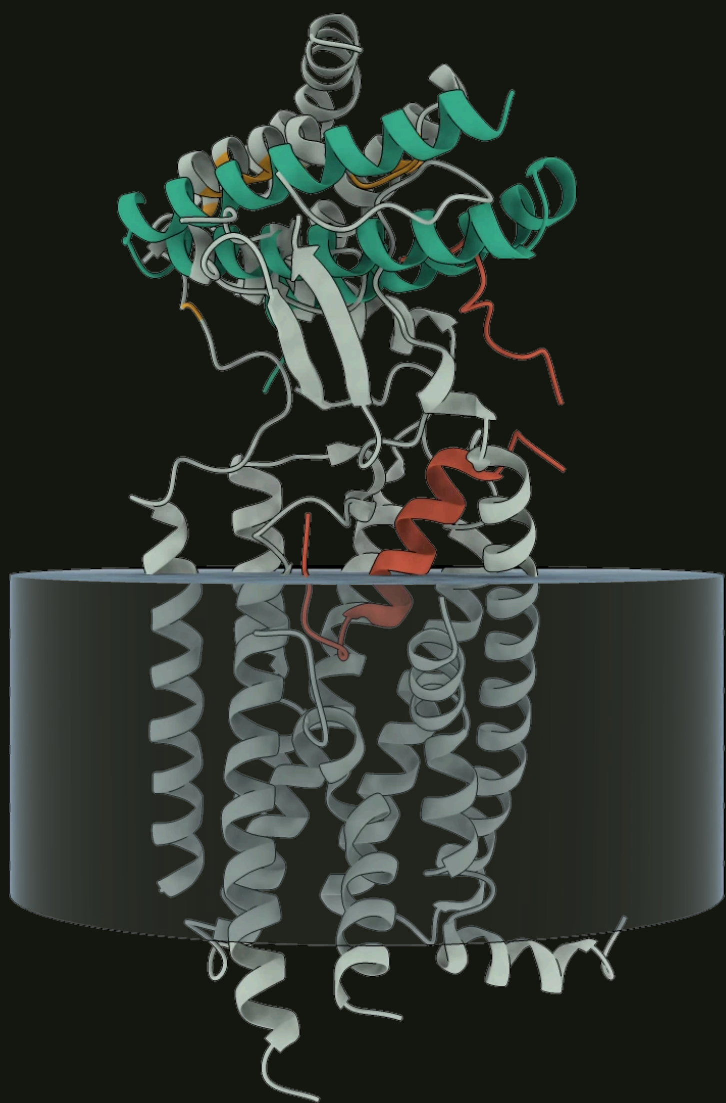
We gave it one sentence about pain.

> “design novel inhibitors for pain receptors that are promising drug targets”

It chose the target, the structure and the epitope, sized the campaign from its own trial, and returned **365 candidate binders** against the receptor behind migraine. **$0.54** of model spend, 9.2 GPU-hours, unattended.



STAGE 1 / NOBODY NAMED A TARGET

Three candidates, ranked by how much evidence stood behind them.

TARGET	EVIDENCE	WHY IT IS DESIGNABLE
CALCRL / RAMP1 **6E3Y (Homo sapiens)**	VALIDATED	Fully extracellular, accessible Class B GPCR ECD / accessory protein interface with proven clinical druggability (validated by approved biologics and small molecules); avoids intracellular delivery hurdles.
NTRK1 / NGF 1WWA (Homo sapiens), 5JZ7 (Homo sapiens)	VALIDATED	High-affinity extracellular interaction directly addressable by designed protein/peptide competitive antagonists.
CDKL5 / CaMKIIα Not found in corpus	PATHWAY INFERRED	Novel intracellular target mechanism selective for peripheral hypersensitivity without systemic receptor blockade.

CHOSEN — AND THE RISK IT FLAGGED ITSELF

CALCRL / RAMP1, the CGRP receptor: the target class of erenumab, an approved migraine antibody, so the mechanism is clinically de-risked before a single design exists.

Potential cross-reactivity with closely related family members (CALCRL/RAMP2 or CALCRL/RAMP3 adrenomedullin receptors).

STAGE 2 / STRUCTURE CHOICE

The best-known structure and the best structure to design on are not the same question.

PDB	RESOLUTION	TARGET LENGTH	SCAFFOLDING CHAINS
6E3Y cited by the literature	3.3 Å	490	4
3N7S chosen	2.1 Å	115	0

THE PIPELINE'S OWN REASON

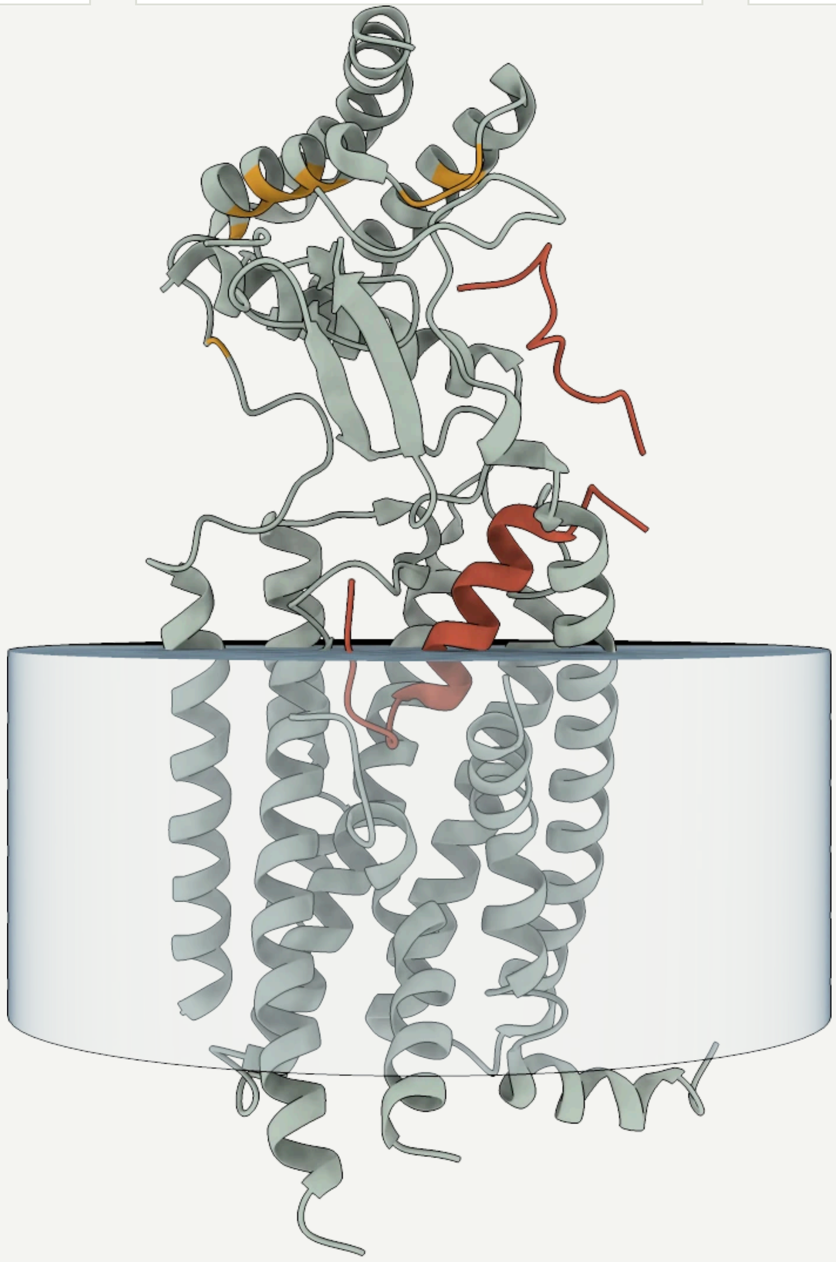
6E3Y carries 4 scaffolding chain(s) (nanobody / G protein / fusion partner) against 0 in 3N7S, at no worse resolution (2.1 Å vs 3.3 Å)

A nanobody, a G protein and a fusion partner are what makes a cryo-EM structure solvable. They are also four extra chains a binder has to be designed around, and none of them exist on a real cell.

STAGE 3 / WHERE A DRUG CAN ACTUALLY REACH

On a membrane receptor, most of the surface is unusable.

10	84	excluded
HOTSPOTS DECLARED	TARGET RESIDUES KEPT	TRANSMEMBRANE FACE



The epitope, on the full-length receptor in the bilayer. Everything below the membrane is unreachable by an injected protein; the transmembrane residues were dropped from the target on both sides.

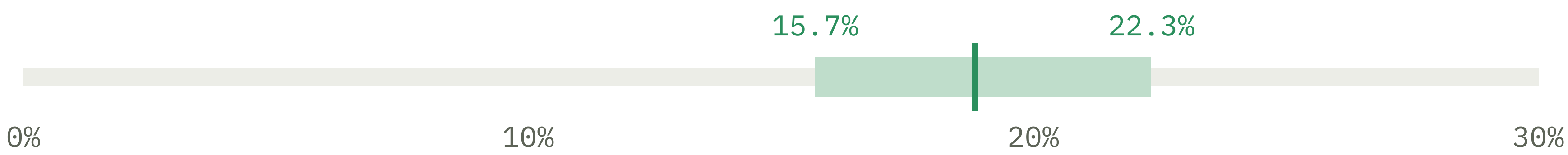
Hotspots: Trp74, Phe83, Trp84, Pro85, Asp90, Phe93, Leu94, His97, Phe101, Ile106.
Priority: F83, W84, P85 — literature-validated contacts on RAMP1.

STAGE 5 / NEVER SCALE STRAIGHT TO PRODUCTION

It measured its own hit rate before spending the GPU budget.

CALIBRATION TRIAL — 99 OF 527 BACKBONES

18.8%



95% Wilson interval. The campaign is sized on the *pessimistic* end, so a lucky trial cannot authorise a run that will not pay for itself.

SCALE UP

VERDICT

0.7 → 0.85

BAR RAISED, NOT LOWERED

3.4 h

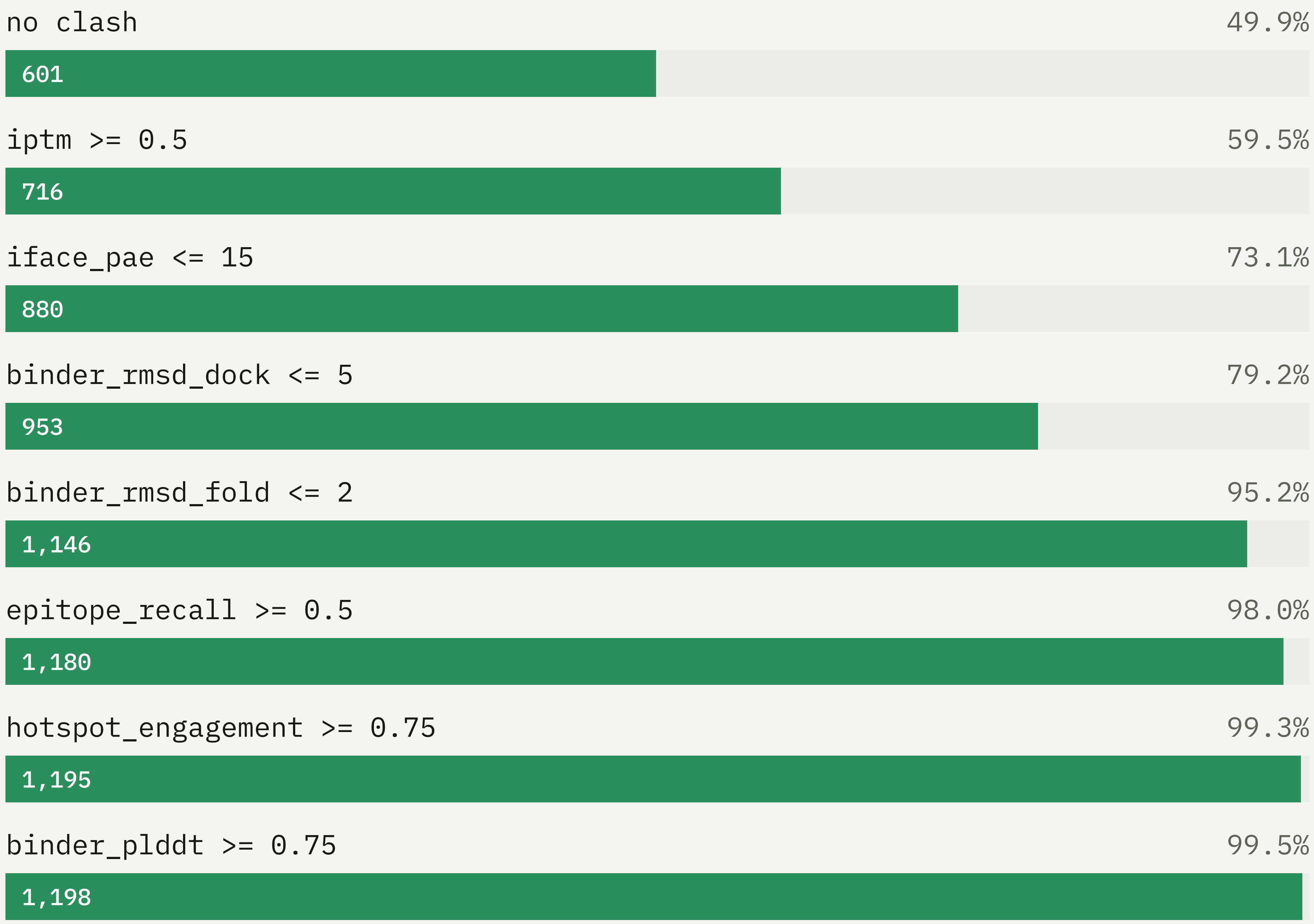
WORST CASE TO HIT TARGET

The target turned out good enough that the pipeline made its own success criterion **stricter** than the one it was given — and still projected the campaign would finish inside budget.

STAGE 6–7 / PRODUCTION AND GATING

Eight independent gates, applied to every refold.

348	1,204	365
BACKBONES	REFOLDS SCORED	SURVIVORS

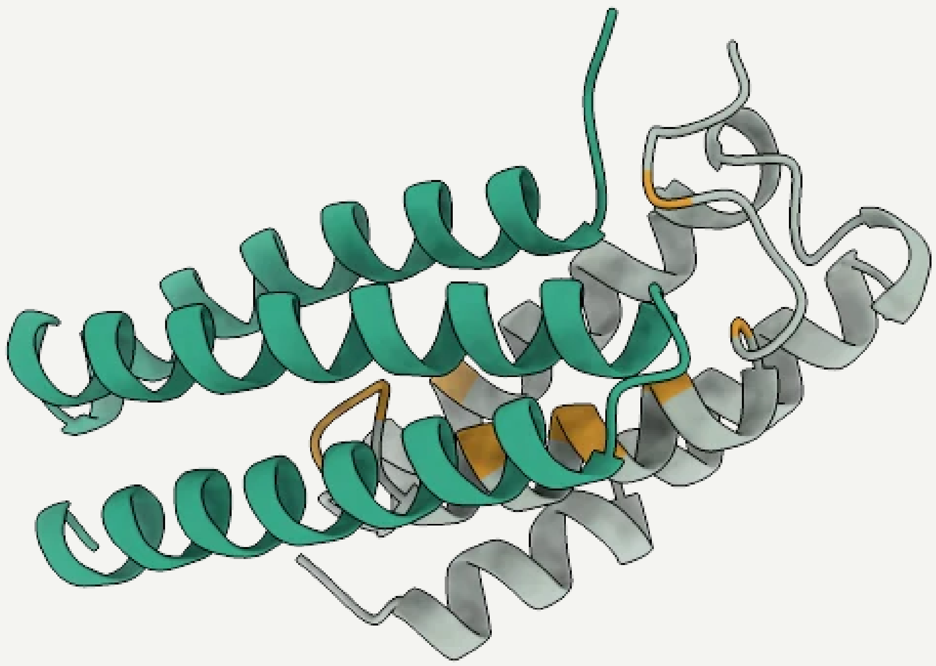


Confidence alone is not enough: a model can be certain about an interface it invented. The geometry gates — docking RMSD, epitope recall, hotspot engagement — are what separate a binder on the right site from a confident one on the wrong one.

STAGE 8 / THE LEAD CANDIDATE

A 83-residue mini-protein, docked to 1.3 Å.

0.908 INTERFACE IPTM	1.28 Å DOCK RMSD
0.722 IPSAE	100% HOTSPOTS ENGAGED



AN INDEPENDENT CHECK THE RUN NEVER SAW

Superposed onto 6E3Y, the full receptor *with its natural agonist bound*, the design overlaps CGRP itself — 122 binder atoms within 4.5 Å, closest approach 0.77 Å. The campaign designed against a peptide-free crystal form and never scored a single design against CGRP.

THE BILL

Unattended, end to end.

$0.54	9.2	4
MODEL SPEND	GPU-HOURS, ONE CARD	STAGES THAT USED AN LLM

Four stages reason: pathway, literature, structure, and the final analysis. Everything between them — trimming, spec building, calibration, sizing, gating, ranking — is deterministic Python, which is why the bill is cents rather than hundreds of dollars and why the same inputs give the same campaign twice.

CITATIONS

14 of 15 DOIs the reasoning stages cited were resolved and verified against the literature. The 1 that did not resolve is named in the run record, not quietly dropped.

Free for any non-commercial use.

WHO	TERMS
Academics, students, non-profits, anyone not doing it for money	Free. Read it, run it, change it, publish with it.
Companies	Needs a commercial licence — ask.

Four campaigns written up end to end — the CGRP receptor, a mesothelioma target found from scratch, a PD-L1 campaign, and a tour of the literature corpus. Every figure is extracted from the run that produced it.

`michauckelmann.github.io/little-protein-tiger`